

Identifying the Needle in the Haystack: A Radar Summary

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Mechanical Engineering



Project – An Idea

- Labelling images is time consuming
 - Especially when the objects are tiny in a sea of water.
- Radar could help identify areas of interests for human reviewers OR automated labelling systems

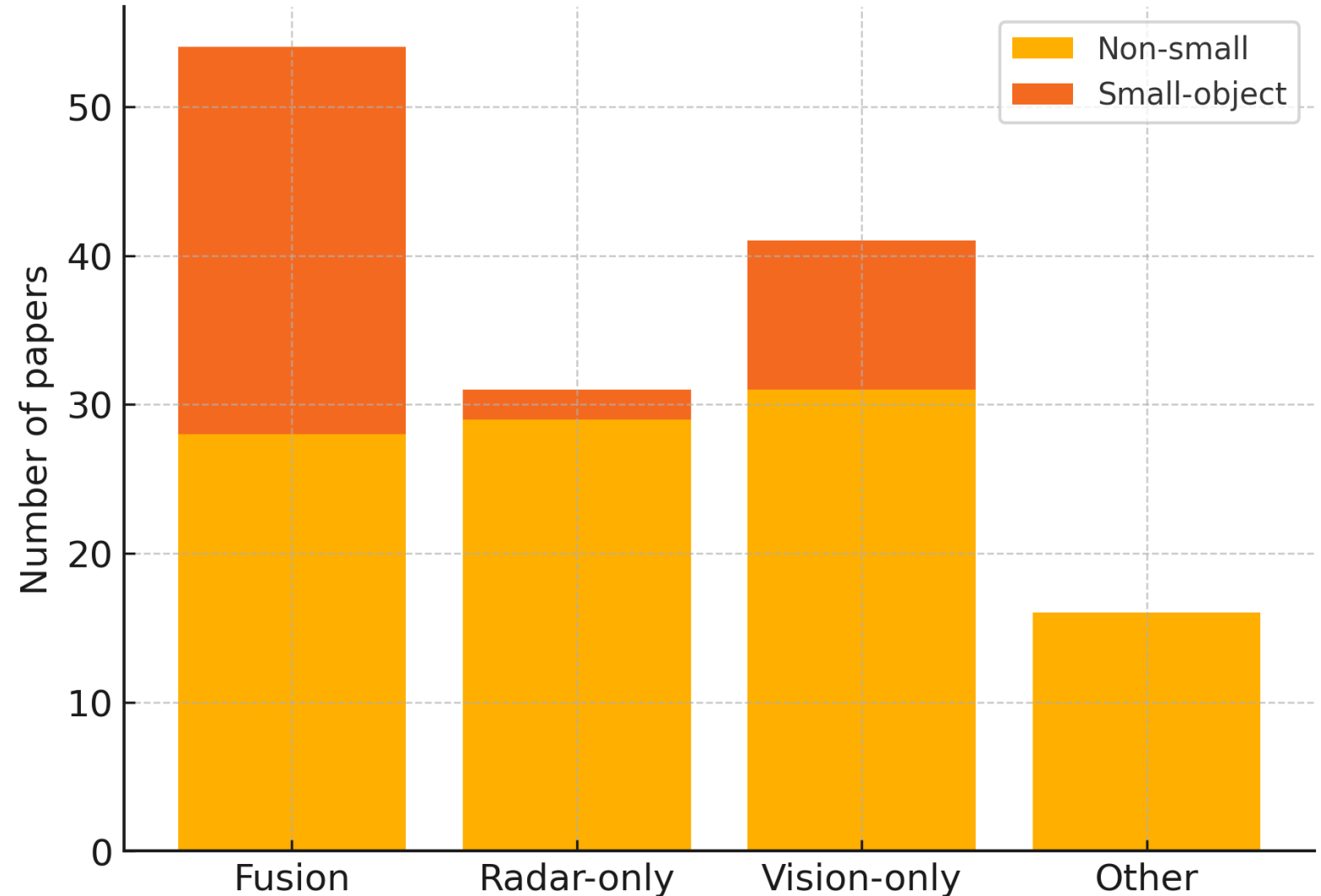


*Not smaller than
32px



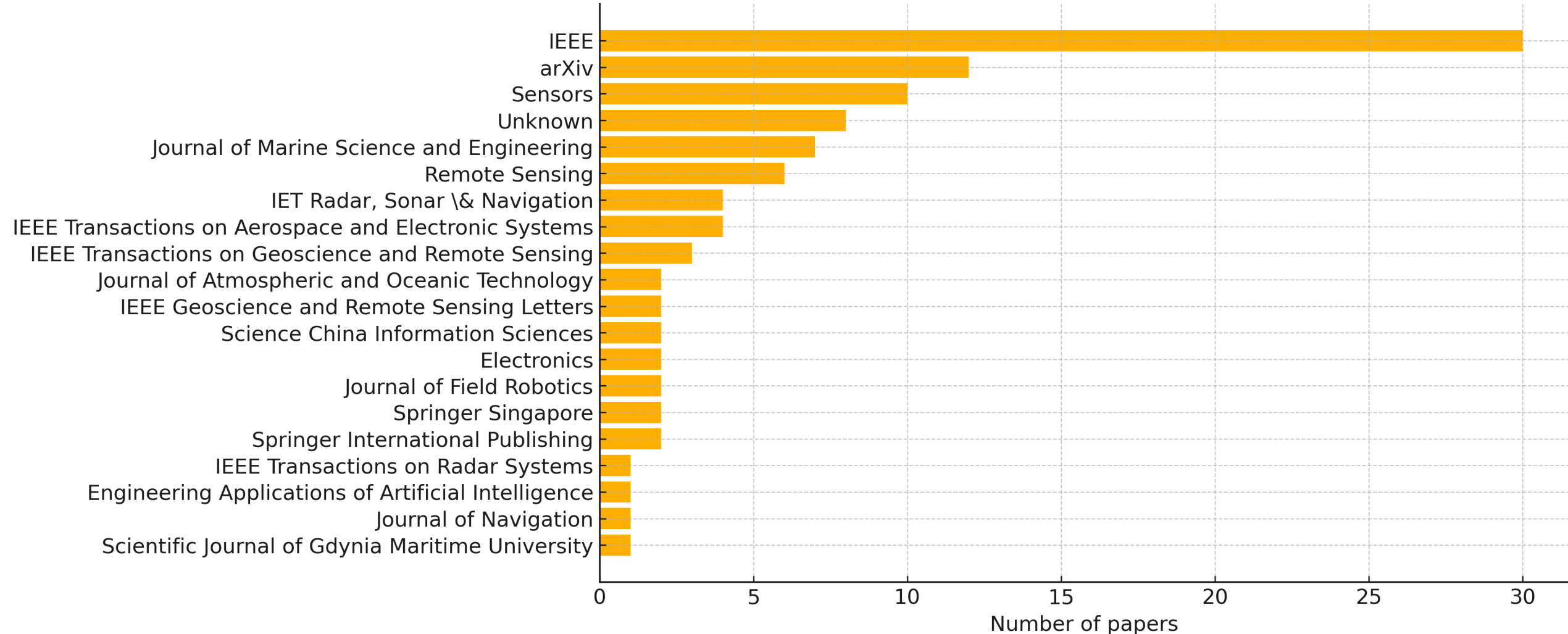
Literature Review – Small-Object Work

- Only 6 % of **radar** papers focus on tiny targets
- **Vision**: 25 %
- **Fusion**: slight minority are small-object



Literature Review – Top Published Venues

Top-20 publication venues (142-paper corpus)



Implementation

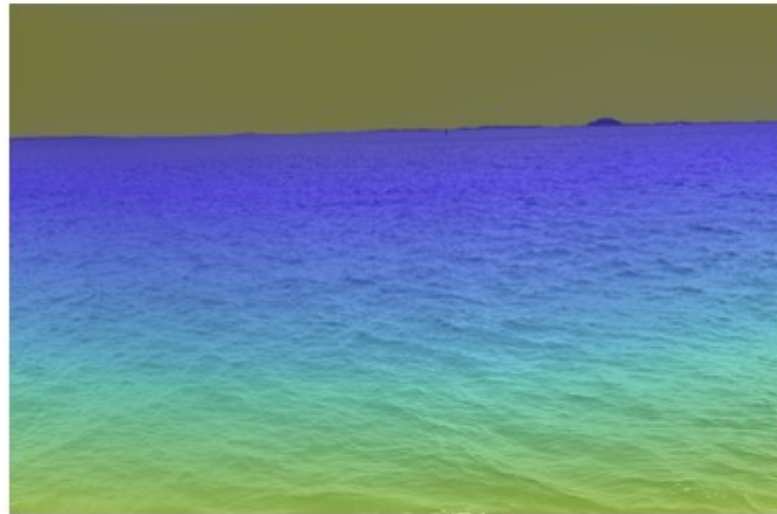
- Segment Image
- Create Depth Map
- Use depth information to overlay radar information over image for human labelers



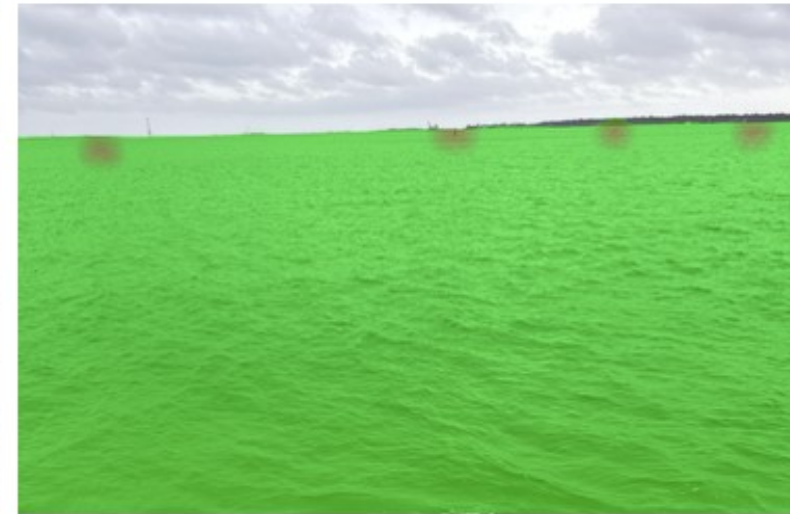
(a) Original image



(b) Segmentation



(c) Depth Map

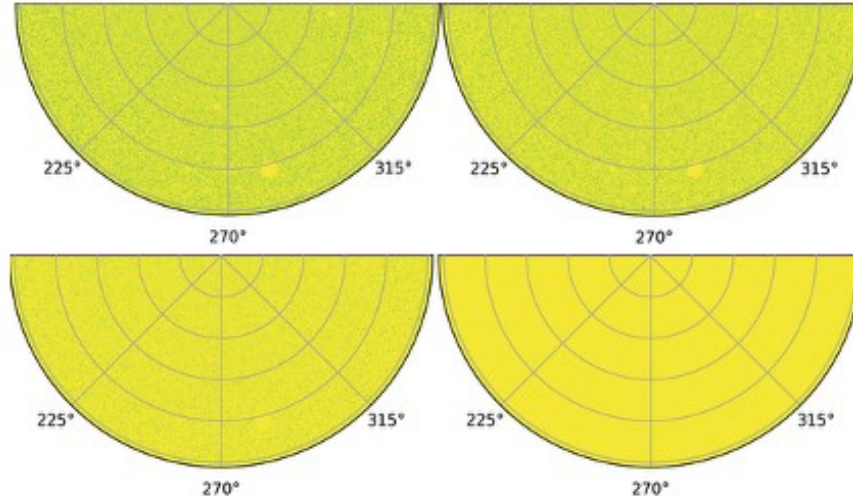


(d) Radar overlay

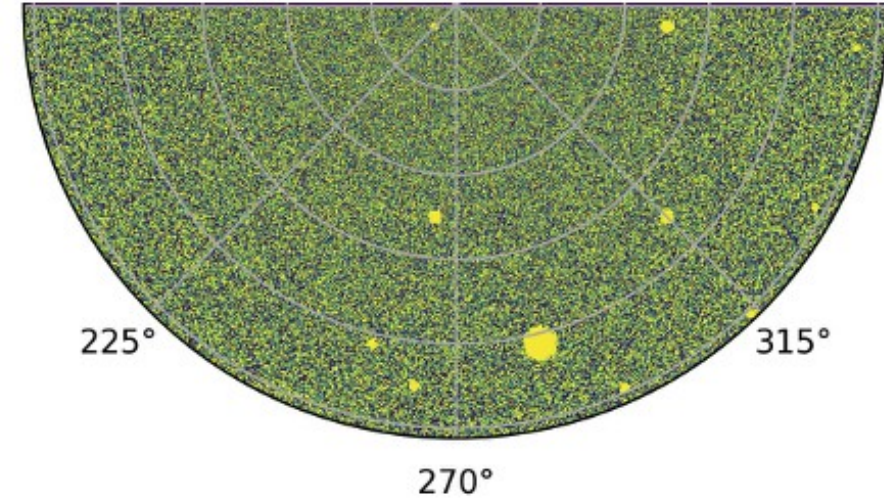
Implementation 2

Use multi-gain to create 'HDR' image of radar data and then apply standard object detection models on this

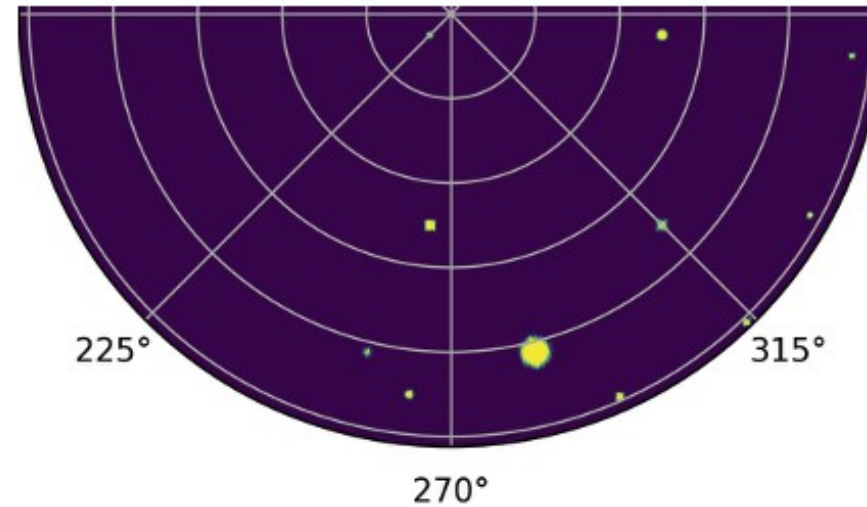
(a) Multi-Gain Capture



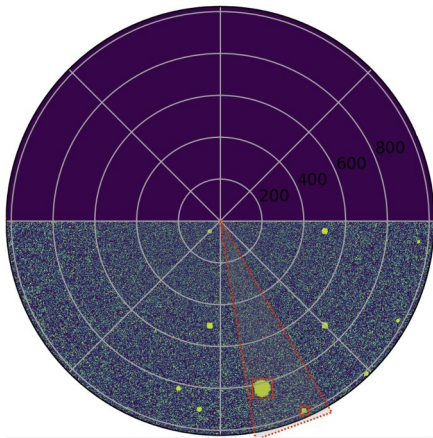
(b) ML Clutter Removal



(c) Ground Truth Objects



(a) Radar Object Detection

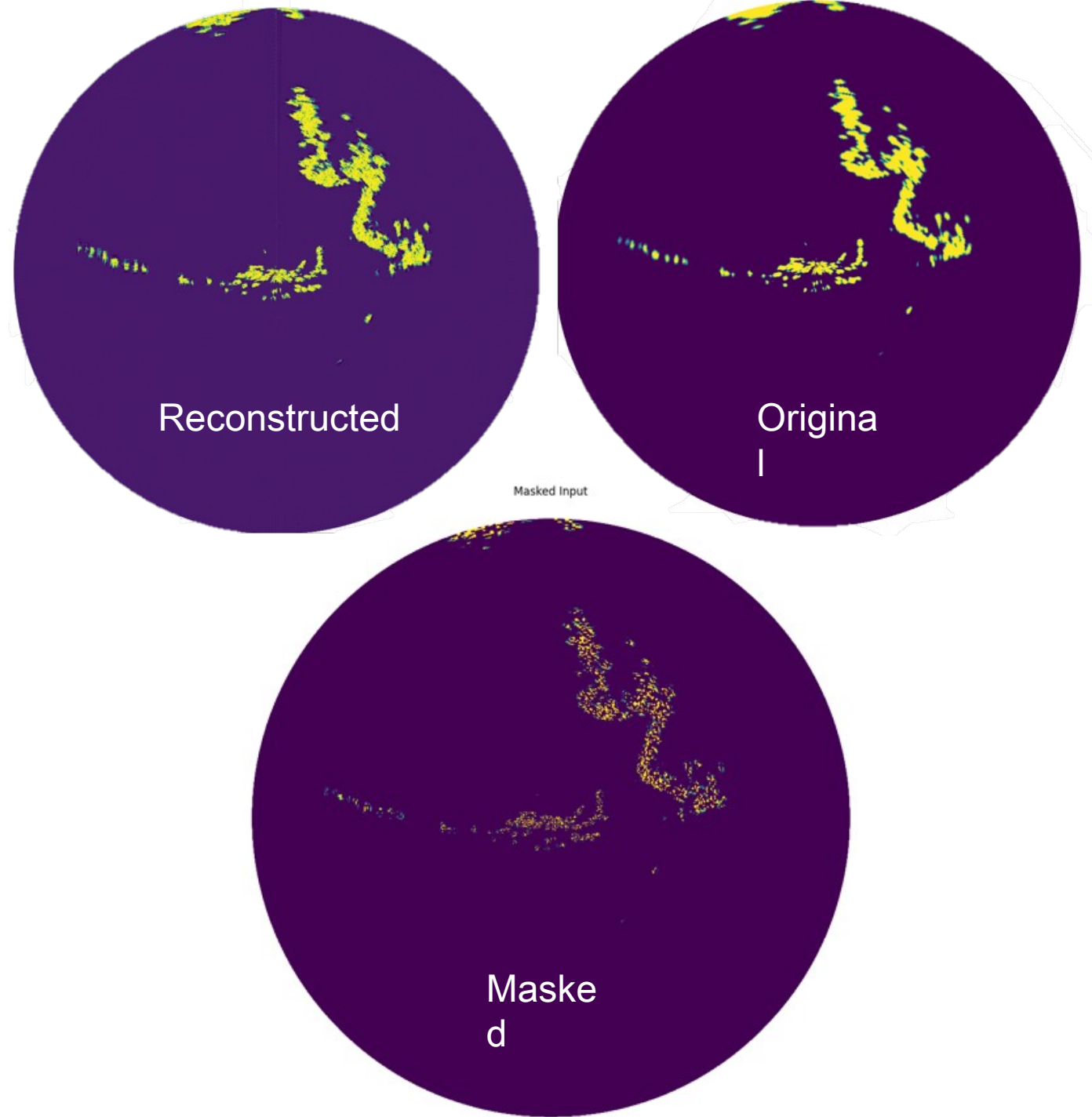


(b) Radar Overlay for Annotator



Implementation 2b

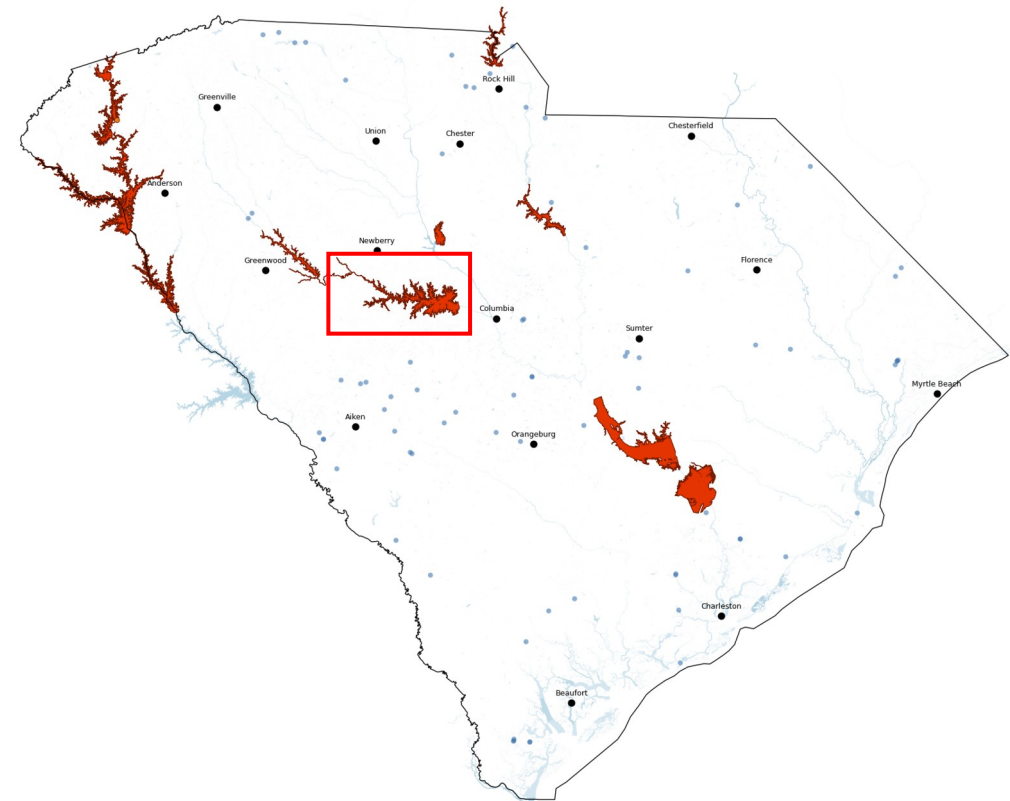
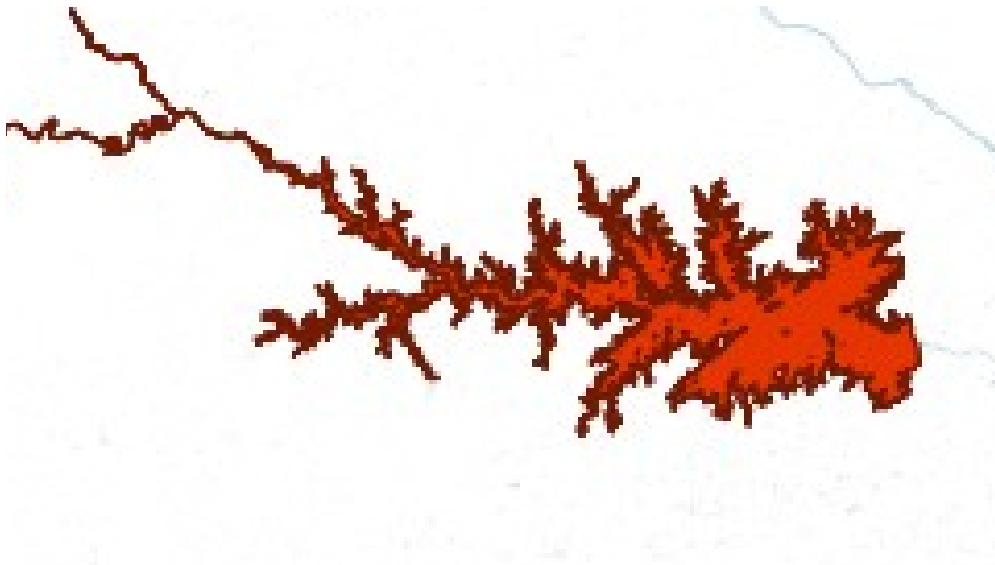
- Self-supervised task
 - No labels, only raw images.
- Feature richness
 - Encoder must infer context, learns shape/texture/semantics.
- Data efficiency downstream
 - Few/one-shot tasks need only a handful of labels once you have the encoder.



Data Collection

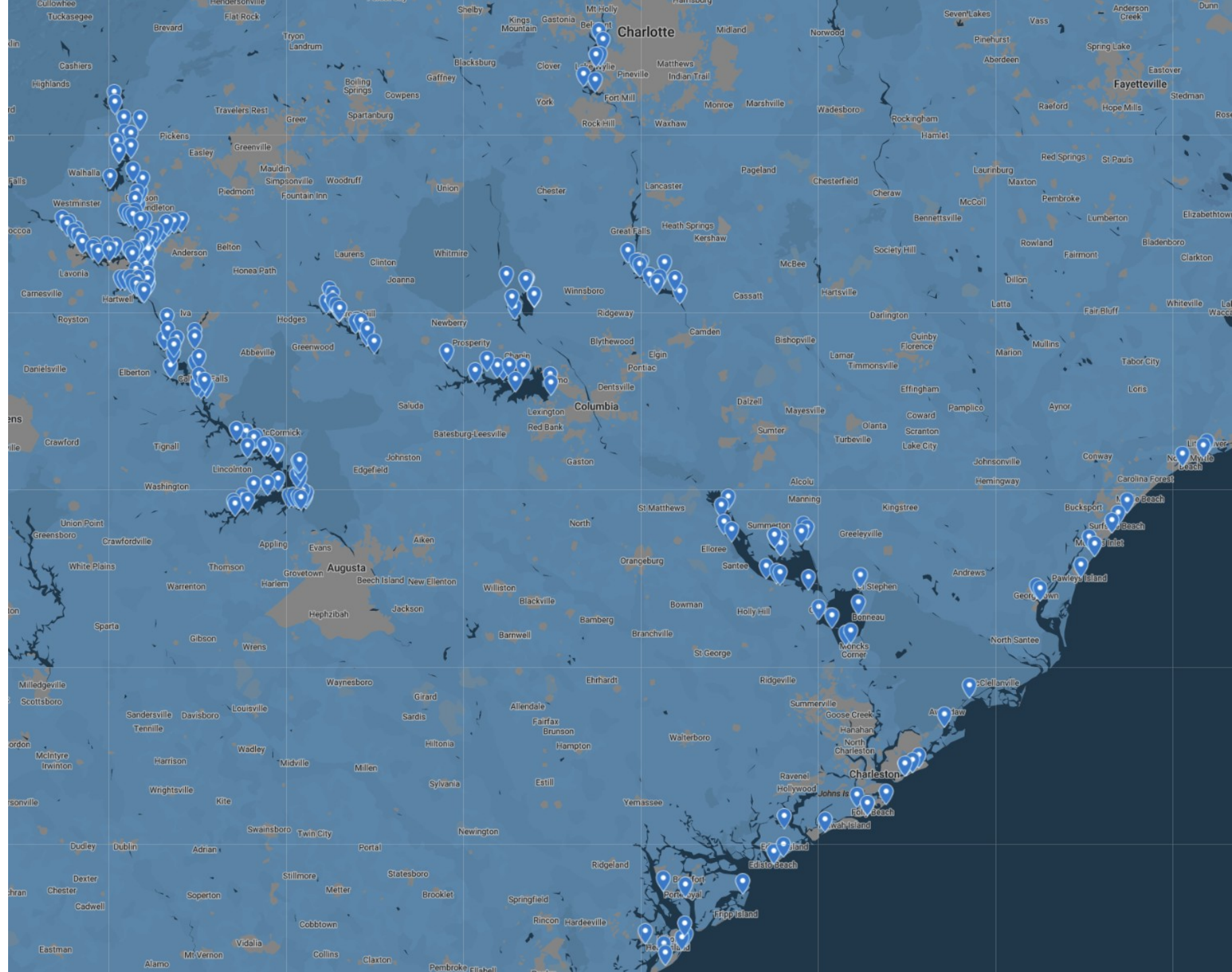
- Collected 50GB(10,000 frames) of data across 6 of Lake Murray's 10 Public Access points as of now.
- Heading, Lat, Lon, and time data are **saved**(along with other info).

This has changed with
'calibration' techniques
mentioned later in PPT!



Data Collection – Where?

- With MAE, a diversity of data is good, else the extracted features will be overfit to a certain location.
- I propose these 198 collection points across SC, possibly adding points in VA at some point.



Capture Rig Computer Casing

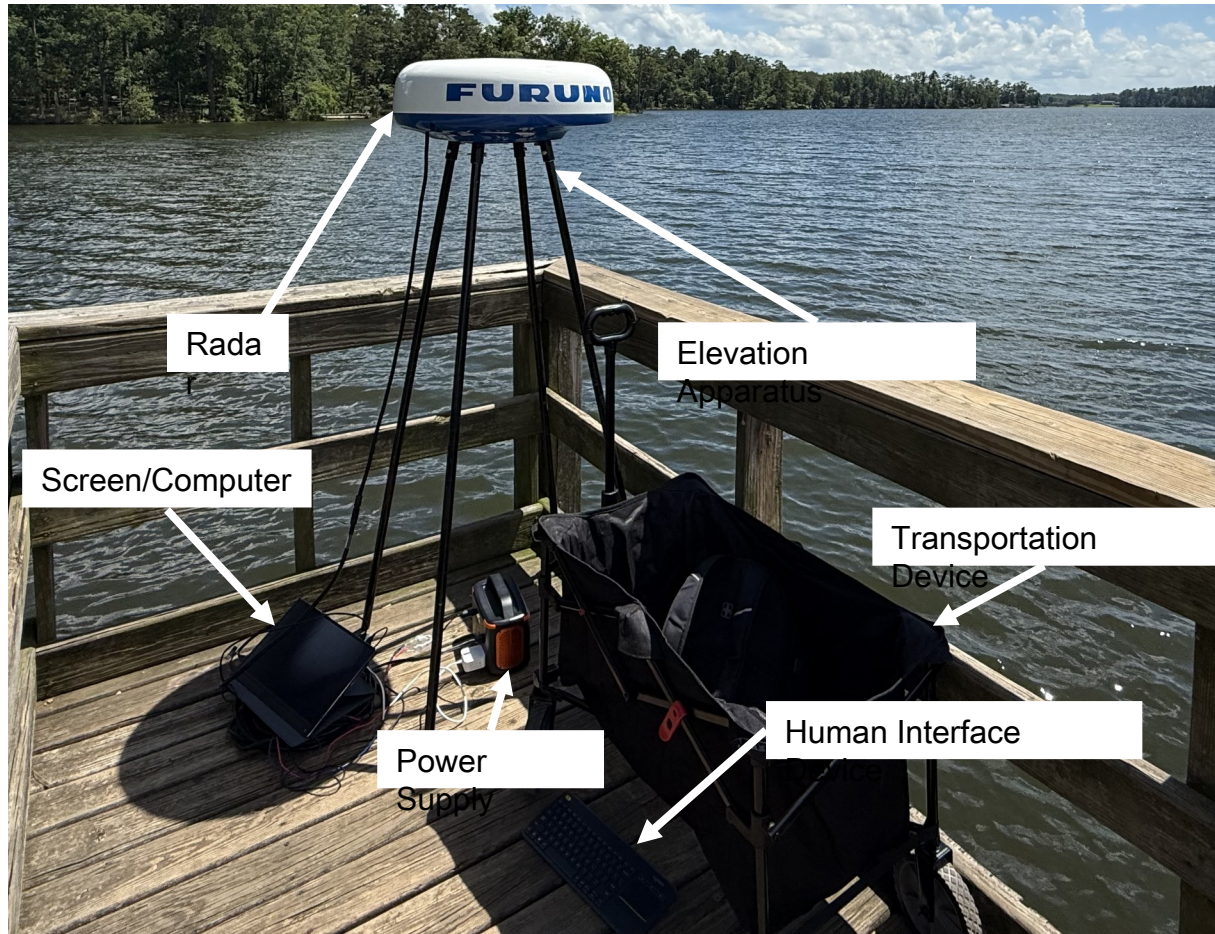


OLD

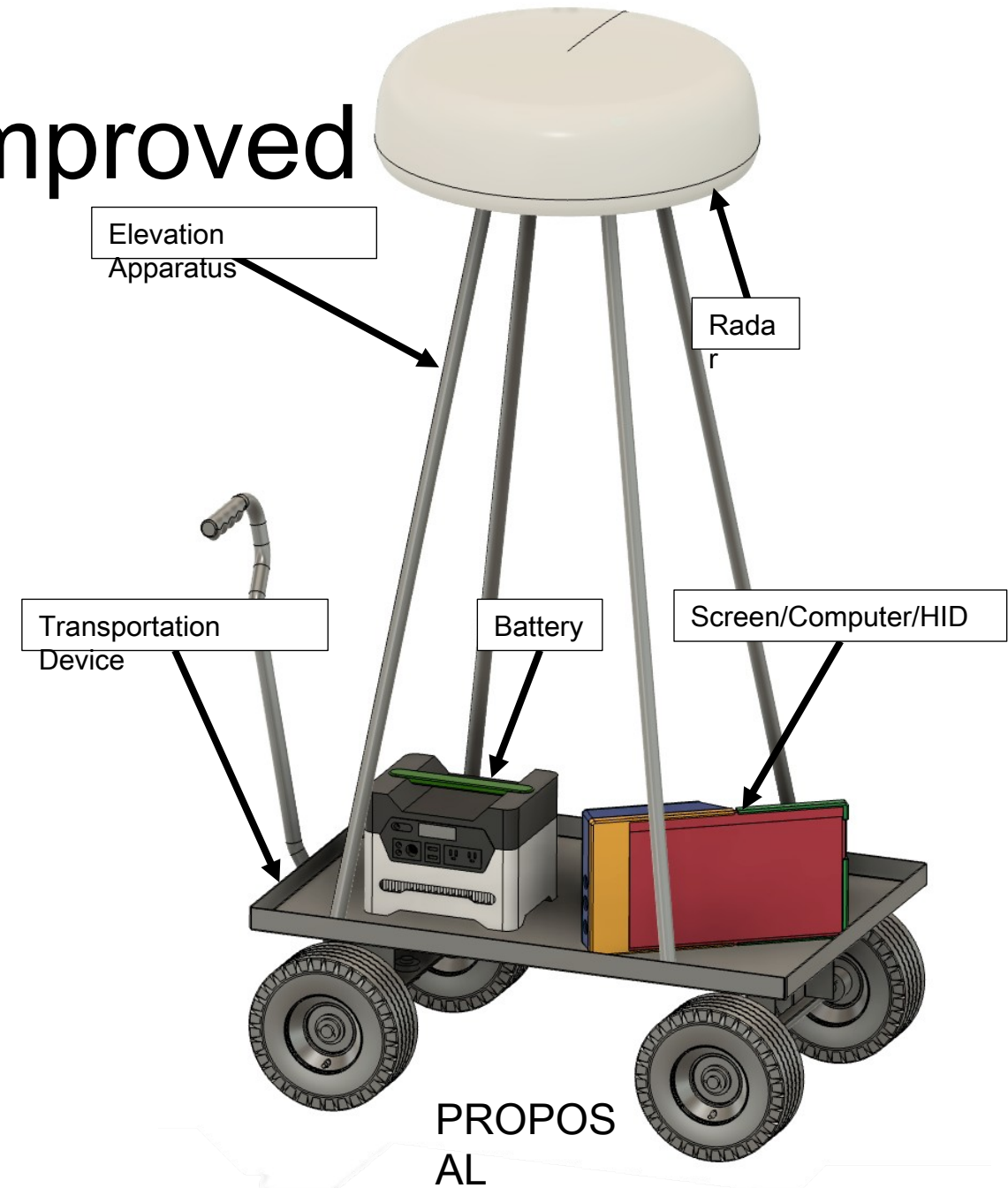


NEW

Capture Rig – Setup Improved



CURRENT





A selection of images.

Calibration – June 30
Macedonia Lutheran Church

A selection of images. Nighttime collection

- Northern Lake Murray
 - Upper Left
- First Successful Field Test: Lake Murray Dam – North Side
 - Lower Left
- Calibration @ Lake Murray Dam – North Side
 - Right



A selection of images.

Equipment needed for
Collection.

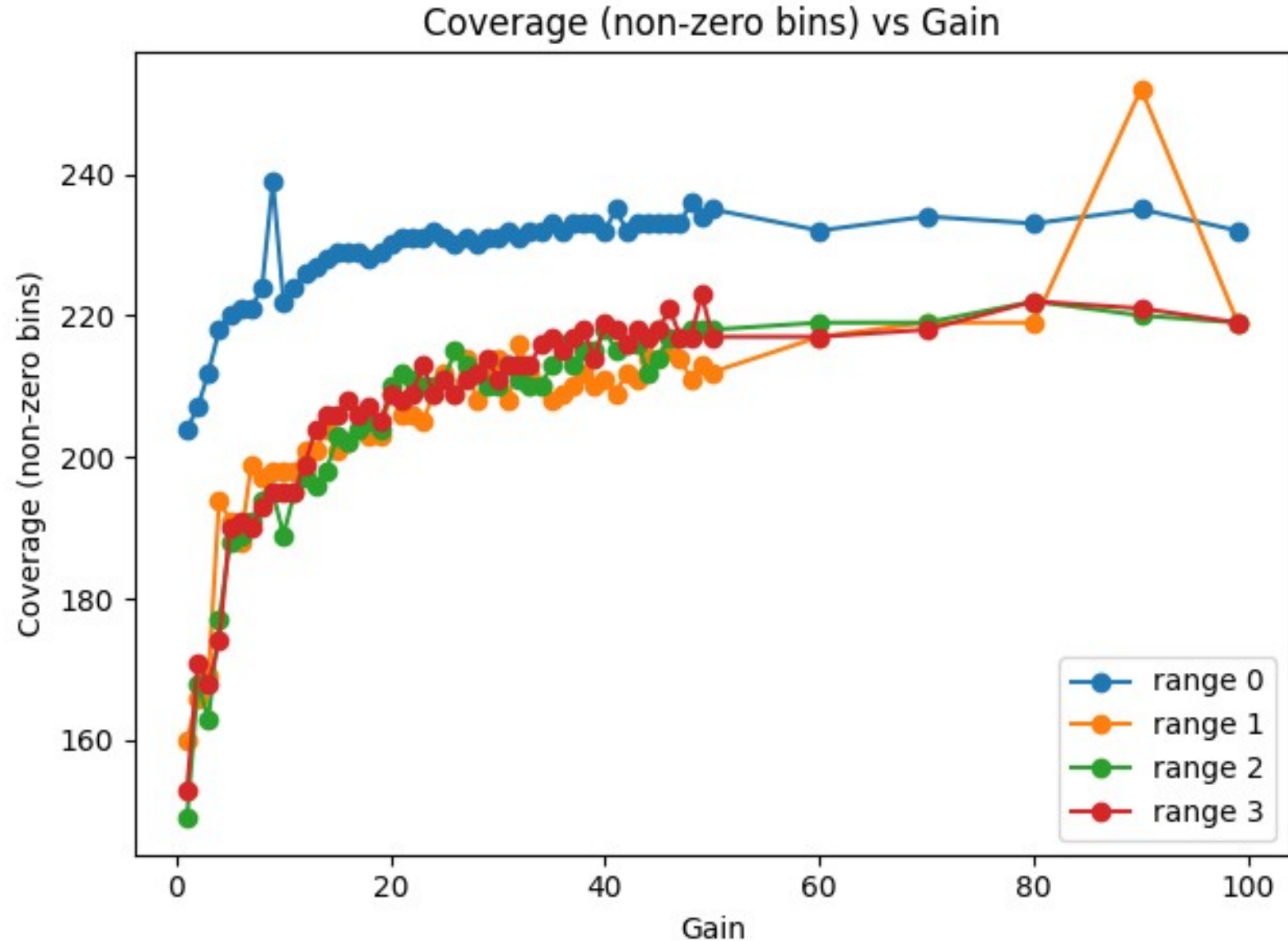




A selection of images.
Nice days for data acquisition

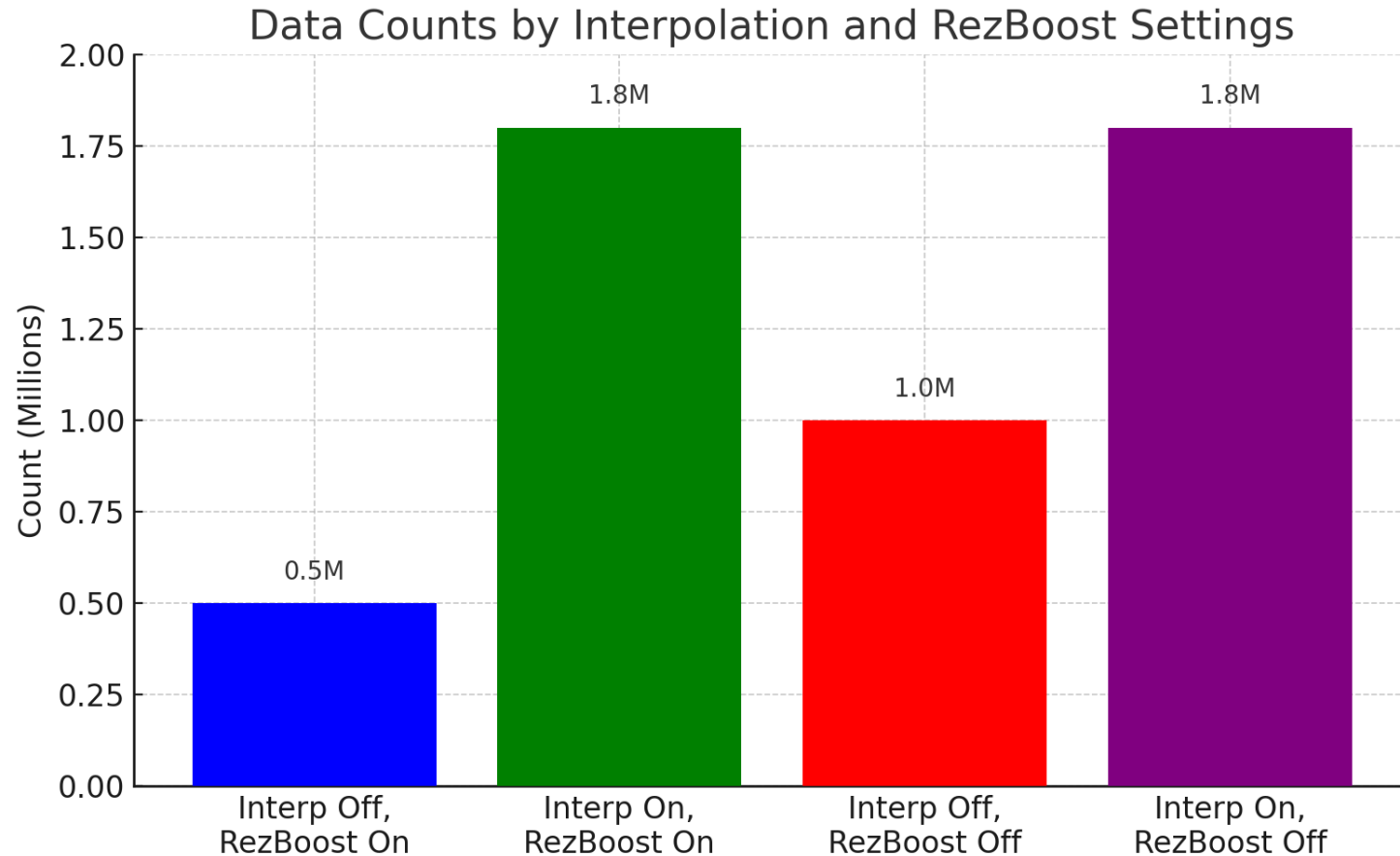
New Info about Furuno Radar

- When collecting data, noticed ~1.1mil samples per sweep
 - ONLY 17 discrete values, basically the same level of accuracy as Simrad
- When interpolation is on, ~1.8mil samples, 253 discrete values
- RezBoost™ (Horizontal Bearing Adjustment)
 - With interpolation on, gets 1.8mil samples
 - With interp off, gets 0.5mil samples



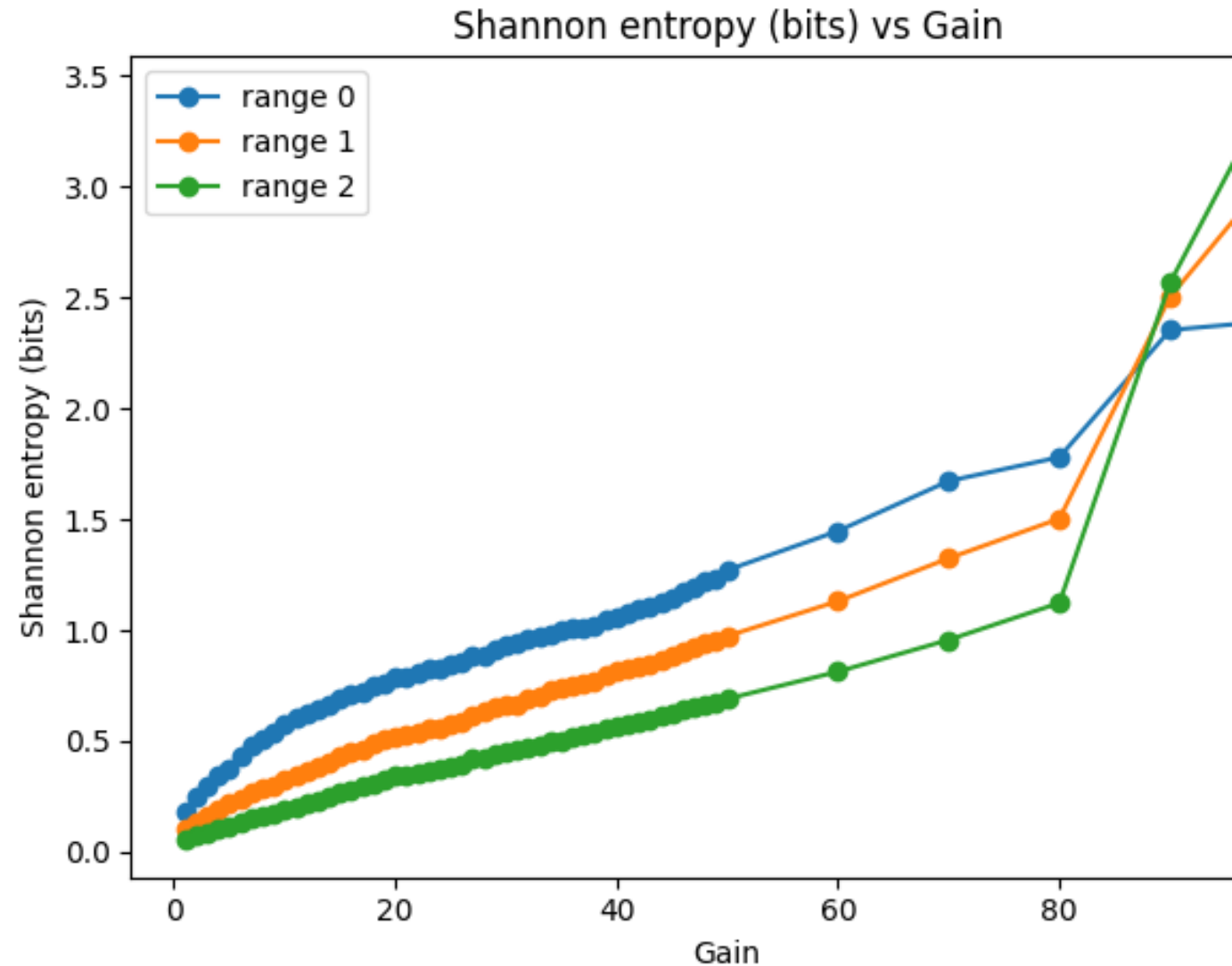
Dual Range-Cell Observations & RezBoost

- **Observation:** Each radial frequently yields two distinct sets of range cells.
- **Initial Assumption:** Extra cells were data corruption artifacts.
- **Test:** Disabling interpolation in RezBoost stripped out all extraneous cells.

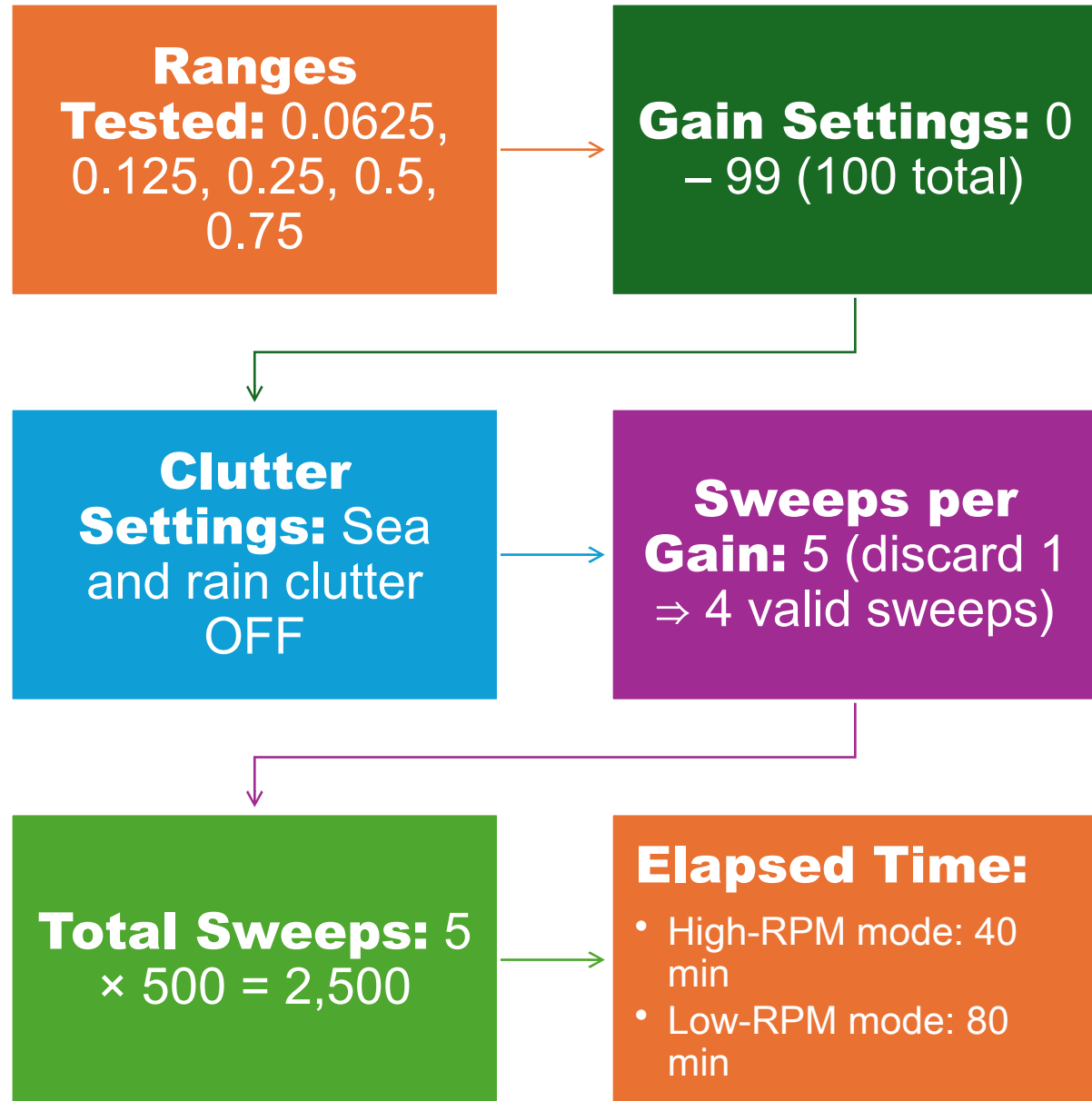
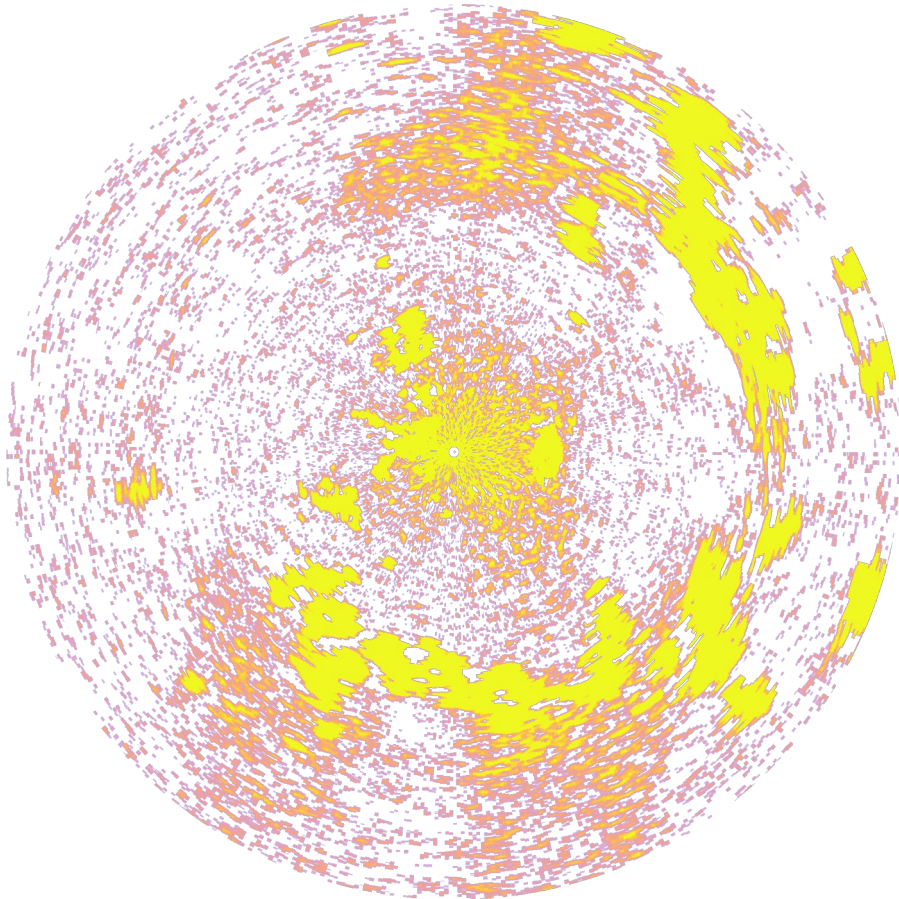


Empirical Gain Selection

- **Objective:** Identify optimal gain settings for model performance
- **Approach:**
 - Gather unlabeled test data across the full gain spectrum
 - Sample at multiple geographic locations
 - Record at several discrete range distances

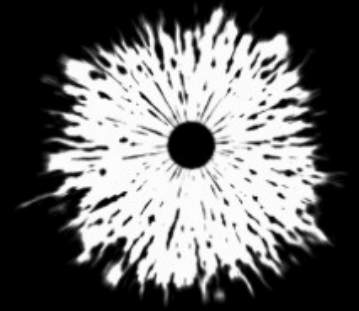


Mini-Calibration Technique



Full-Calibration Technique

- **Interpolation:** On / Off (2 settings)
- **RezBoost:** All four modes (4 settings)
- **Ranges:** 0.0625, 0.125, 0.25, 0.5, 0.75 (5 settings)
- **Gain:** 0 – 99 (100 settings)
- **Clutter:** Sea and rain OFF
- **Sweeps per Gain:** 5 (discard 1 $\Rightarrow$ 4 valid sweeps)
- **Total Sweeps:** $2 \times 4 \times 5 \times 500 = 20,000$
- **Elapsed Time:**
 - High-RPM mode: 320 min (5 hr 20 min)
 - Low-RPM mode: 640 min (10 hr 40 min)



Rezboost+Inter
p

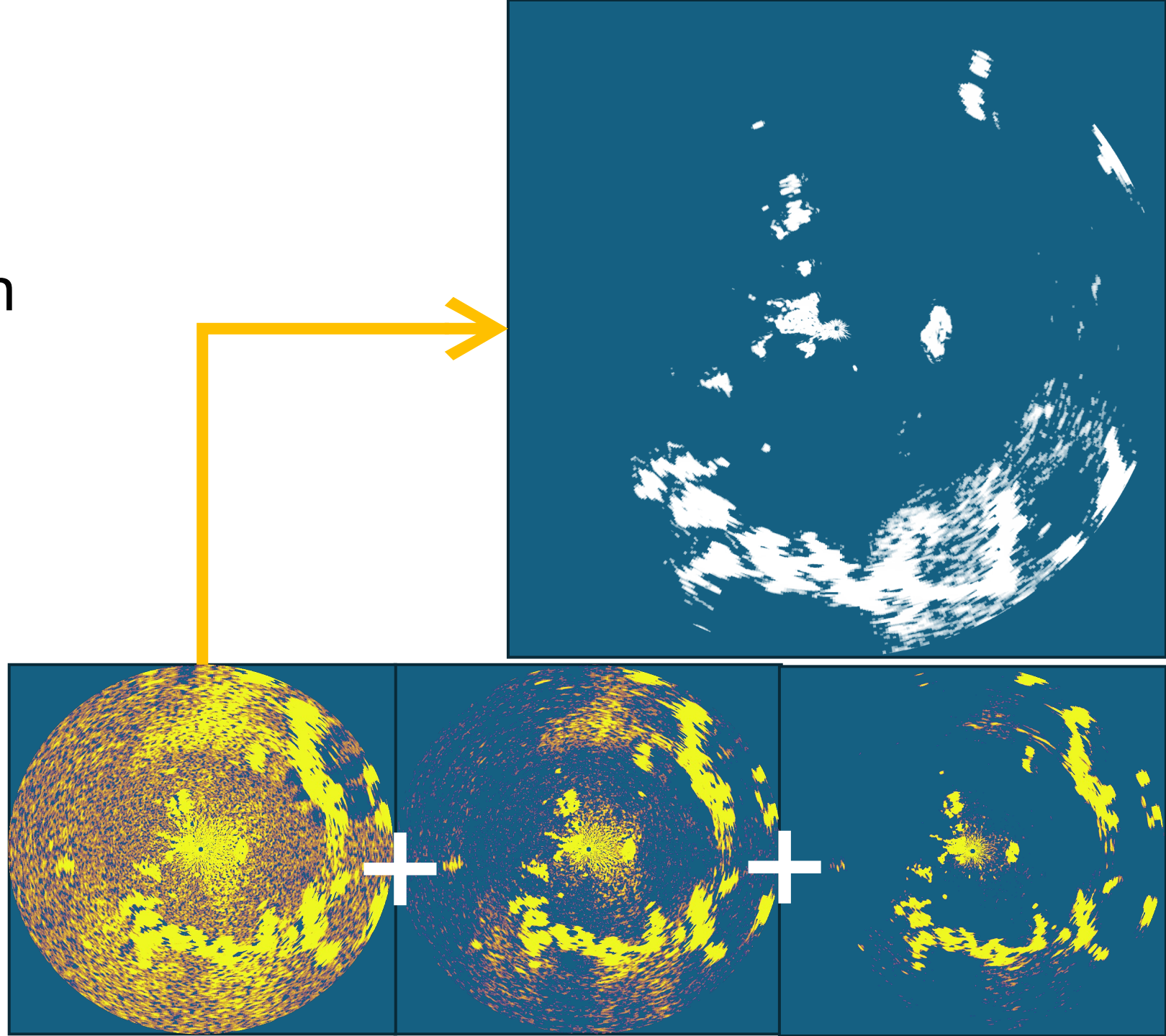


RAW+Inter
p

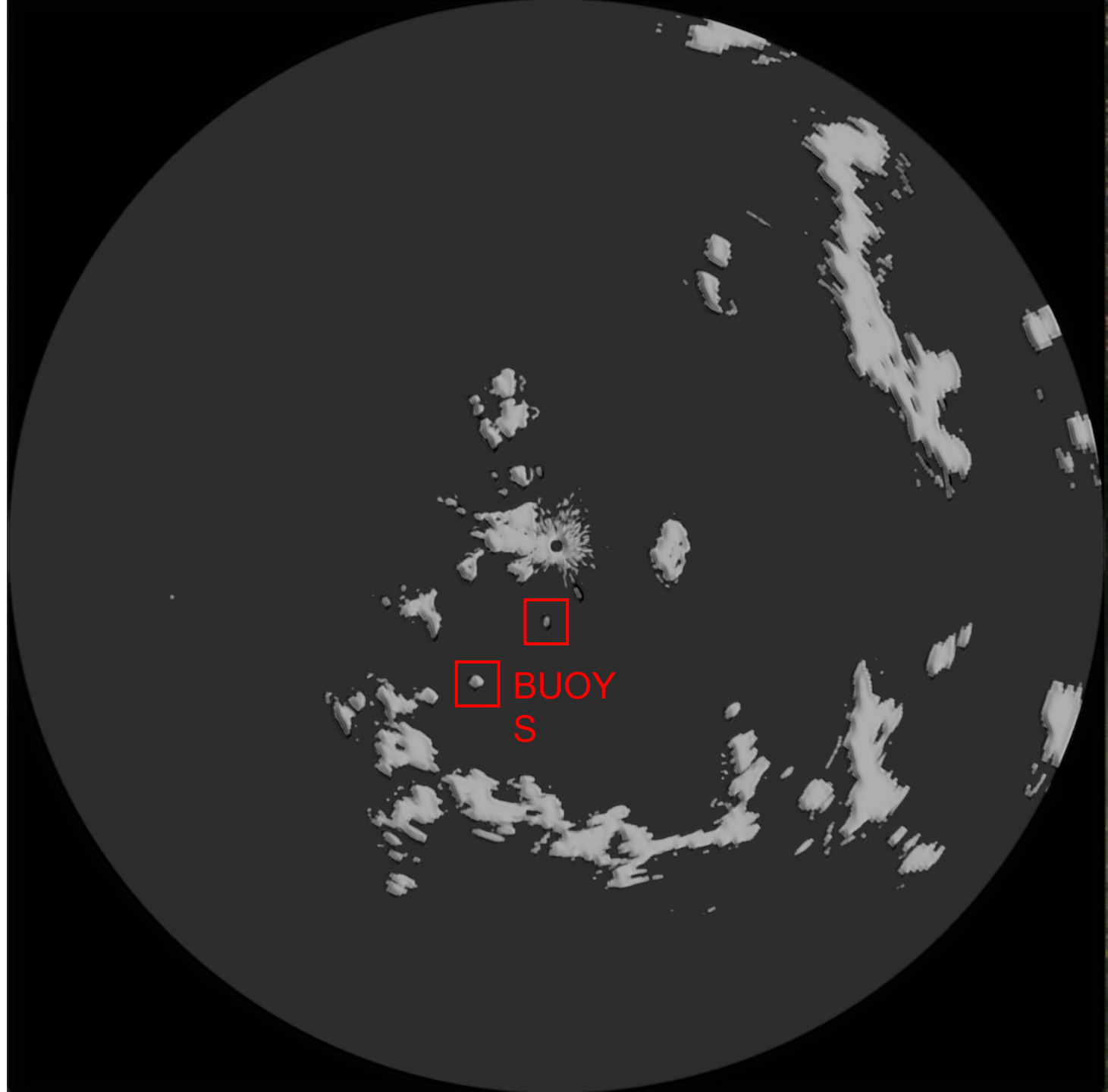
HDR Attempt

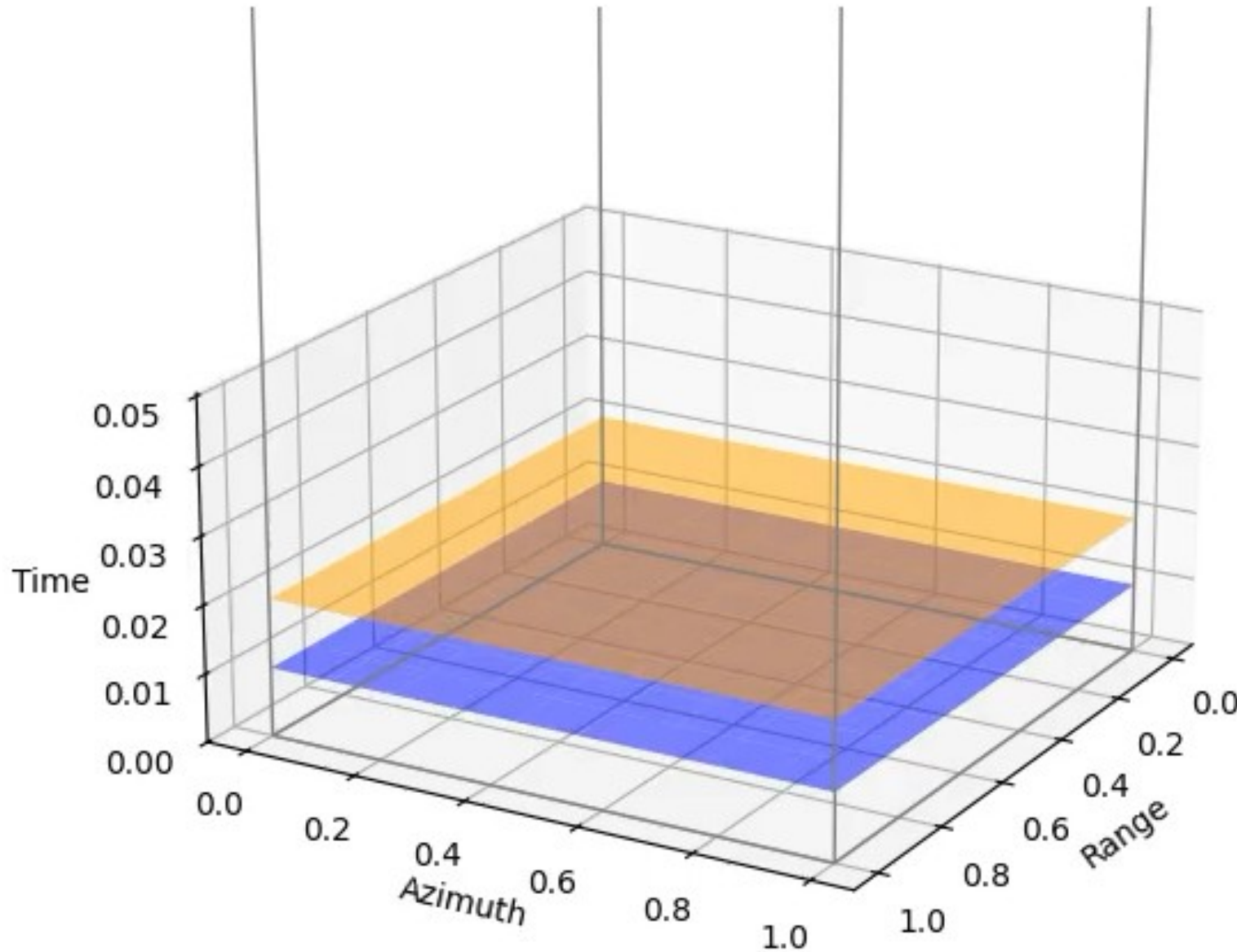
Used multiple fusion algorithms

- MTI
- Wavelets
- OpenCV's Merten
- BM3D
- N2V



Attempt #5
at HDR
N2V
(Noise2Voi
d)





Radar Data

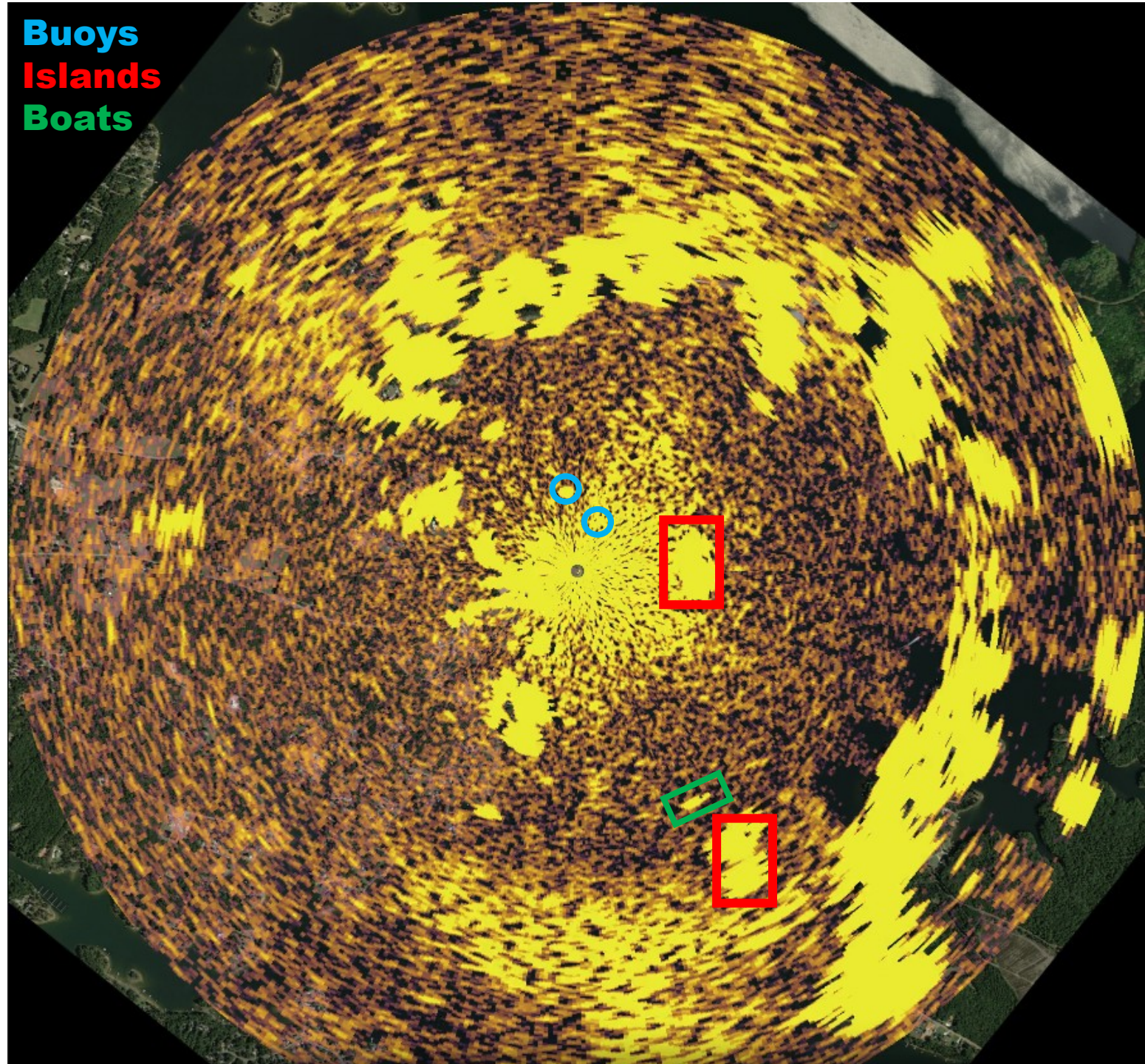
Self Supervised 4D Masked Autoencoding for Dual-Pulse Marine Radar

“I hate labelling data”

– Everyone at some point

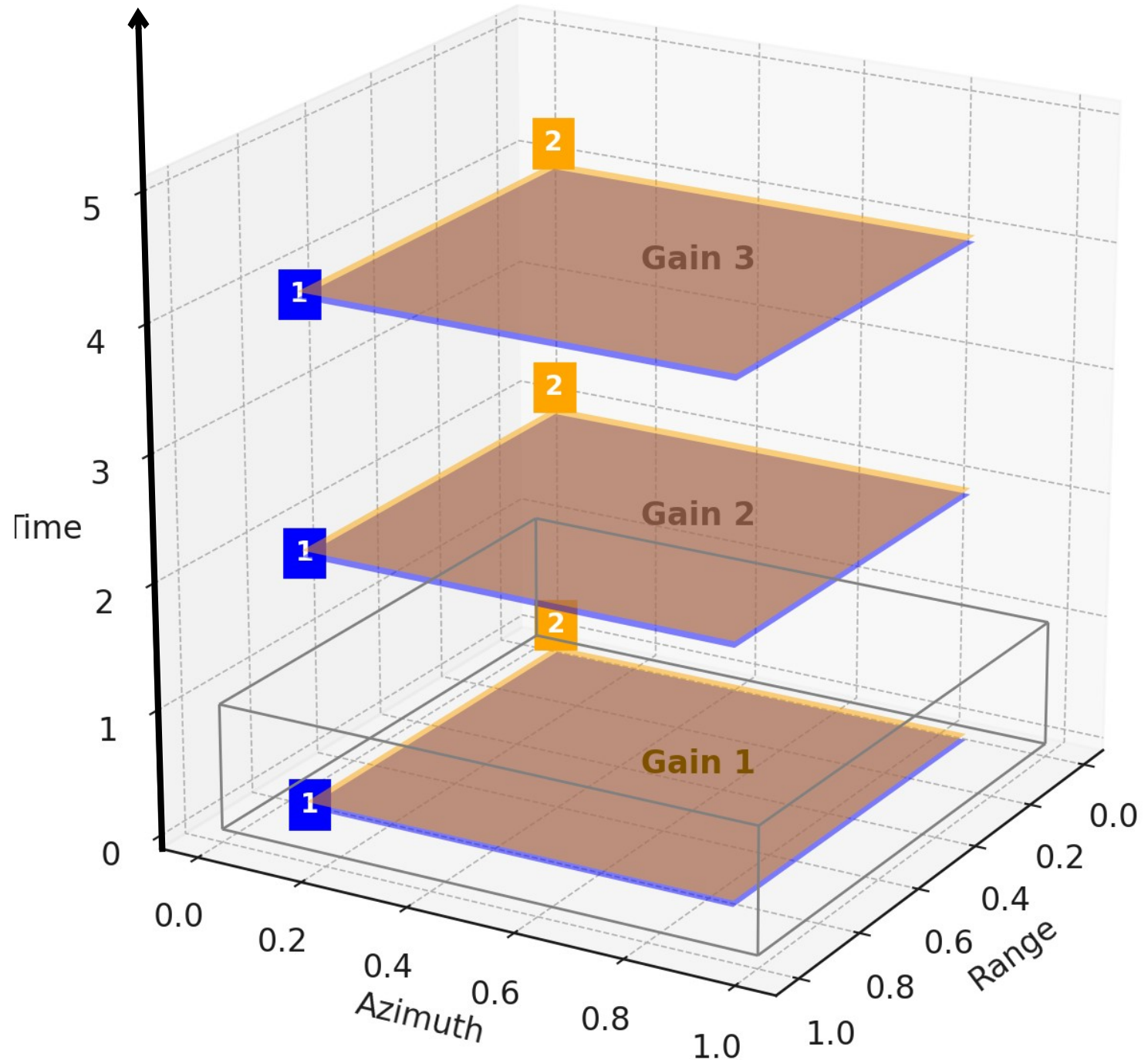
Radar needs new learning tricks.

- Sparse labels are costly
- Stationary + moving targets blend
- 80 dB dynamic-range swings between pulses
- Standard CFAR ignores context



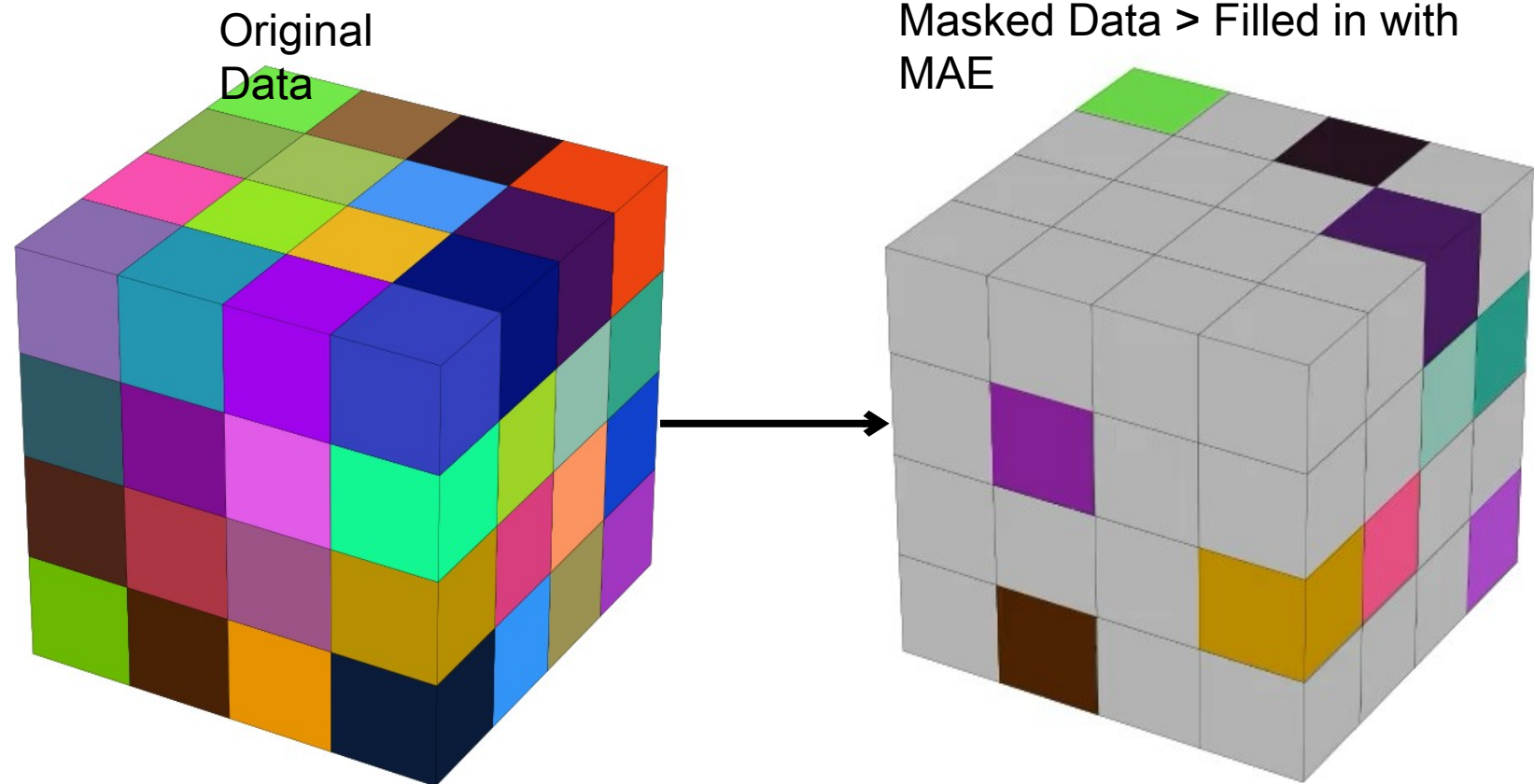
The Data Tensor: 4D Radar Cube

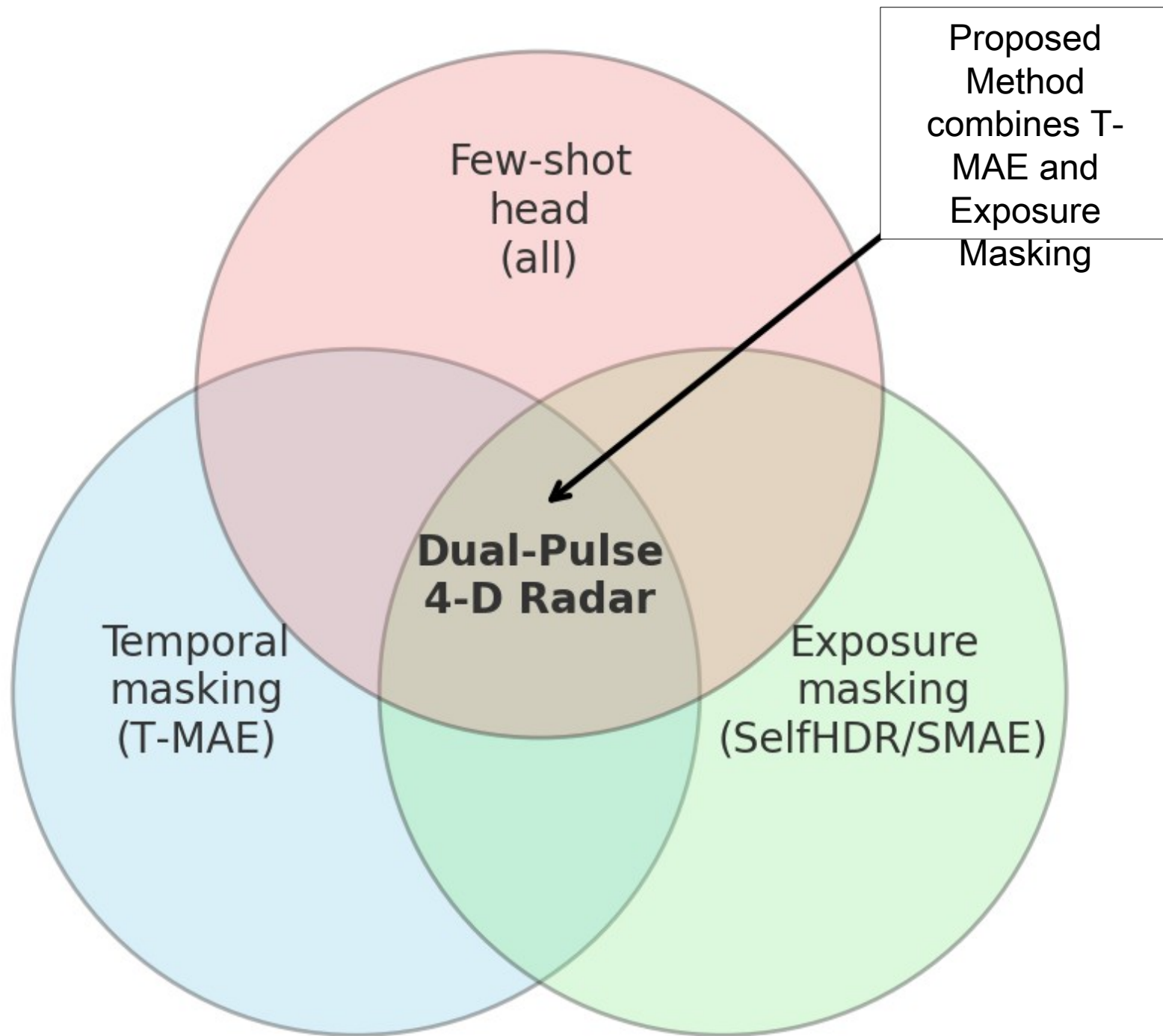
- Axes: Range × Azimuth × Time/Gain × Pulse
- Pulses 1 and 2 are taken at same power level
- Used to:
 - Estimate noise
 - Reduce clutter
 - Improve interpolation



Step 1: Temporal Masked Autoencoder(T-MAE)

- Randomly mask 75 % voxels *per frame*
- Condition on previous frame => learns motion prior
- Loss = L2 + SSIM on reconstruction
- Init weights: T-MAE (ECCV '24)



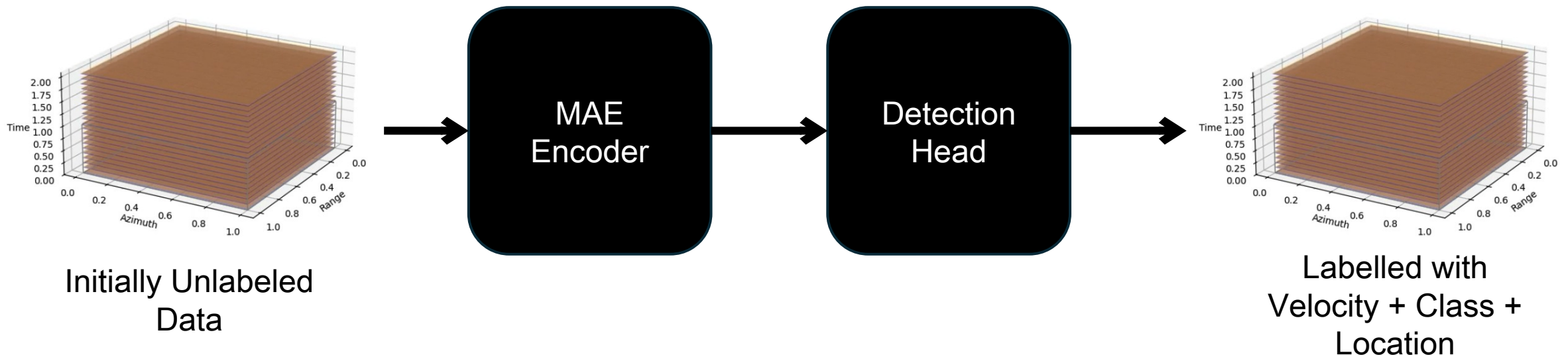


Step 2: Exposure-Aware Masking

- Inspired by SelfHDR (ICLR'24) & SMAE (CVPR'23)
- Treat clipped voxels in Pulse-1 as *masks*
- Fuse low/high pulses by reconstruction-loss

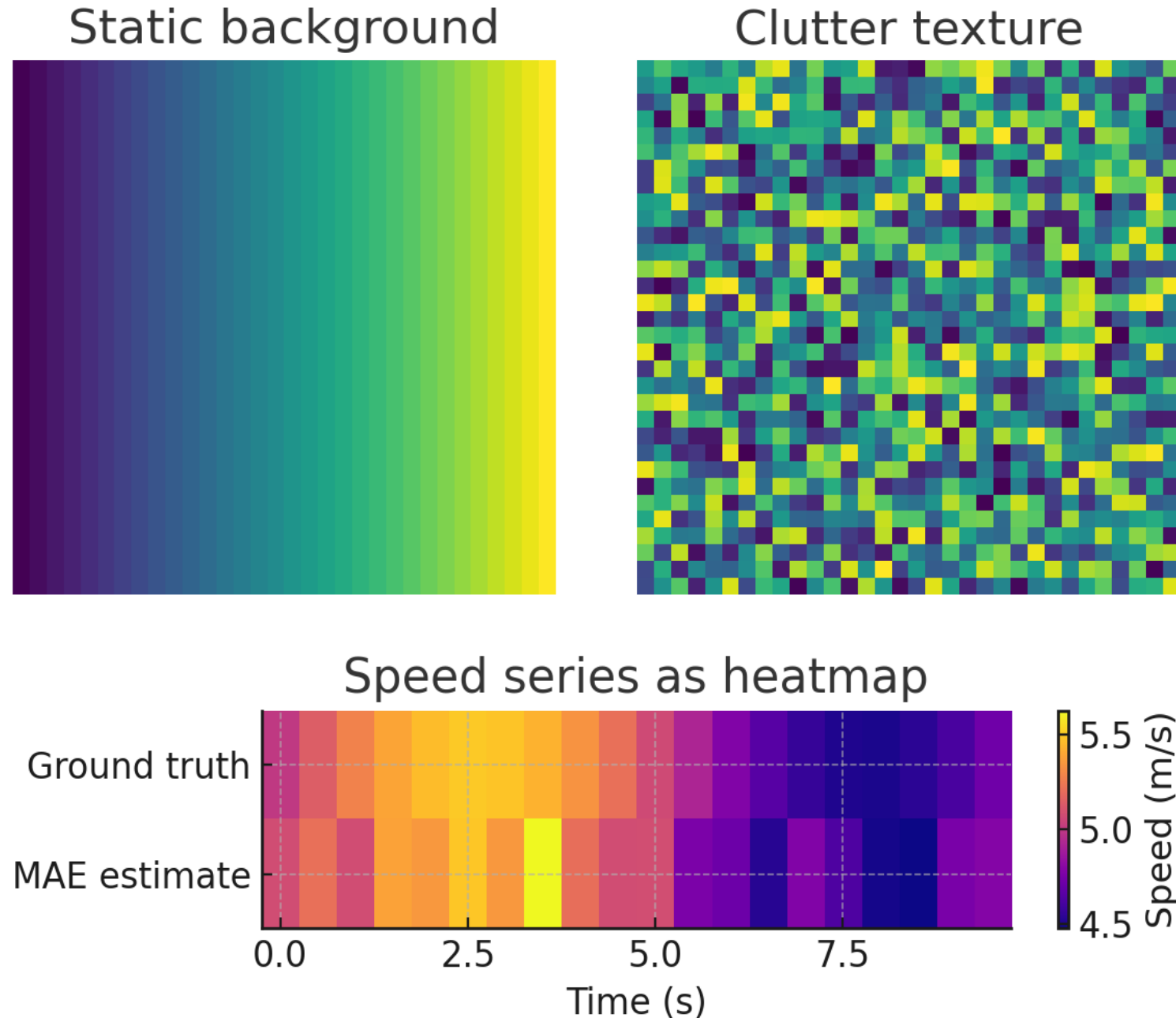
Few-Shot 3-D Detection Head

- Freeze MAE encoder
- Add lightweight 3-D YOLO/CenterHead
- Train on very few labelled frames
- Output: class + 3-D box + velocity



What the Model Learns in Self Supervision

- Stationary objects (background prior)
- Moving targets & speed
- Dynamic-range robustness (dual-pulse)
- Clutter texture statistics



Other Examples in Literature (skip slide)

Paper (venue+year)	Sensor	Core idea	Reported gain	Motion	HDR	Few-shot	Link
T-MAE (ECCV 2024)	LiDAR point-cloud sequence	Temporal Masked Autoencoder uses previous frame to reconstruct highly-masked current frame	+9.2 mAPH on Waymo with only 5 % labels	✓		✓	
RiCL (arXiv 2024)	Range-Doppler radar heat-maps	Instance contrastive pre-training exploits temporal views	Reaches full-supervised mAP with 20 % labels	✓		✓	(arxiv.org)
4-D Radar Scene-Flow (RA-L 2022)	4-D automotive radar pts	Self-supervised scene-flow network separates static vs moving clutter	Robust scene-flow with zero labels (supports motion segmentation)	✓			(arxiv.org)
Self-Sup Velocity Estim. (IV 2022)	Automotive radar	Bootstraps velocity from its own predictions over time	Matches supervised velocity error without velocity GT	✓			(arxiv.org)
SelfHDR (ICLR 2024)	Multi-exposure camera images	Self-supervised HDR reconstruction fuses exposures	+1.58 dB PSNR over prior self-sup baseline		✓		(github.com)
SMAE (CVPR 2023)	Multi-exposure camera images	Saturation-Aware MAE, then few-shot de-ghosting	Beats 2nd-best by +0.15 dB PSNR-L in 5-shot HDR		✓	✓	(openaccess.thecvf.com)

Timeline - Tentative

- Weeks 1-4: Collecting Data, prepping Radar, tuning Gain parameters, validating basic Idea
- Weeks 5-?: improving model, making tweaks, labelling some data...
- Weeks ?-??: Training detection head. Preferably before end of '25

